POWER BI CASE STUDY

Retail Analytics Transformation

Building a Unified Sales Intelligence Platform

Industry: Retail | Tool: Microsoft Power BI
Focus: Data Integration, Modeling & Dashboard Design

1. Business Context & Problem Statement

1.1 Company Overview

RetailMax is a fast-growing consumer retail company operating across Tier 1, Tier 2, and Tier 3 cities in India. Over the past two years, the company has expanded to over 50 pin codes, introduced multiple product categories, and scaled its customer base significantly. However, rapid growth has also brought operational complexity that the business is struggling to manage with its current reporting infrastructure.

1.2 The Business Problem

Leadership at RetailMax faces a critical challenge: despite having access to large volumes of data, they cannot answer basic strategic questions in a timely manner. The core issues are:

- Data is scattered across multiple disconnected files — city tier mapping, pin code geographies, product catalogs, and raw transaction records exist in isolation.
- There is no single, reliable source of truth for revenue, cancellations, or customer activity.
- Reports are generated manually in spreadsheets, leading to inconsistencies and delays.
- Leadership cannot distinguish weekday vs. weekend sales patterns or track weekly trends — information critical for staffing, promotions, and inventory decisions.
- City names in the data contain inconsistencies (country suffixes, alternate spellings) that corrupt geographic analysis.

> **Business Goal**
> Design and deploy a unified Power BI analytics model that integrates all retail data sources, enforces data quality, and delivers an interactive management dashboard — enabling leadership to make fast, confident decisions on revenue, cancellations, customer trends, and product performance.

1.3 Stakeholders & Their Needs

Stakeholder	Role / Decision Need	Key Questions to Answer
CEO / MD	Strategic growth & market expansion	Which city tiers and zones drive the most revenue?
VP Finance	Revenue accuracy & cancellation control	What is net realized revenue after cancellations?
Sales Head	Regional & product performance	Which products and cities are underperforming?
Operations Manager	Demand forecasting & staffing	How do weekday vs. weekend volumes differ?
Analytics Team	Data integrity & reporting pipeline	Is the data model clean, consistent, and scalable?

2. Data Landscape & Integration Strategy

2.1 Source Data Inventory

The analytics team identified four core data sources that together describe RetailMax's complete retail operation. None of these files were previously linked, leading to fragmented and often contradictory reporting.

Data File	Contents	Key Column	Links To
CityTier	Maps each city to a market tier (Tier 1/2/3) and Zone (North/South/East/West)	City	PinCode-Geo
PinCode-Geo	Maps pin codes to cities and regions for geographic drill-down	City, PinCode	Sales Transactions
Product Mapping	Categorizes product codes into categories and sub-categories	Product_Code	Sales Transactions
Sales Transactions	Core fact table: orders, units, cancellations, revenue by date and customer	City, Product_Code, Order_Date	All dimension tables

2.2 Loading & Connecting Data in Power BI

The first action the analytics team takes is importing all four data sources into Power BI Desktop using "Get Data". Each file is loaded into Power Query for inspection before it reaches the data model. The team ensures:

- Headers are correctly promoted from the first row for all tables — preventing Power BI from auto-generating generic column names (Column1, Column2, etc.) that would break downstream calculations.
- Data types are explicitly assigned: dates as Date, units as Whole Number, city names as Text.
- All four tables are loaded as separate queries and held in the data model for relationship building.

3. Data Quality & Cleansing

3.1 Why Data Quality Matters Here

Incomplete records in a retail model don't just cause cosmetic issues — they corrupt aggregated KPIs. A transaction row missing Zone would prevent correct regional revenue totals. A row missing CityTier would exclude sales from tier-based comparisons. The analytics team's mandate is zero tolerance for incomplete dimensional attributes.

3.2 Removing Incomplete Records

Inside Power Query, the team filters the Sales Transactions table to eliminate any row where the Zone or CityTier fields are null or empty. This is a deliberate business decision: unclassified transactions cannot be meaningfully assigned to any region or market segment, so they are excluded from the model.

Business Impact

By enforcing completeness on Zone and CityTier, every revenue figure in every report can be confidently attributed to a geographic segment. Finance and regional managers can trust that their numbers are complete and accurate — not understated due to missing mappings.

3.3 Standardizing City Names

During data profiling, the team discovers that city names in the transactions file contain a country suffix — for example, “Mumbai, India” instead of simply “Mumbai”. This mismatch prevents the City column from joining correctly with the CityTier and PinCode-Geo dimension tables, which store clean city names without the country part.

The resolution is executed in Power Query:

- The original City column is preserved as City_Old to maintain an audit trail and allow rollback if needed.
- A new City column is created by extracting only the text before the comma, effectively stripping the country suffix.
- All downstream joins now use this cleaned City column, ensuring full relationship integrity across the model.

4. Building the Data Model

4.1 Establishing Relationships

With clean data in place, the analytics team moves to Power BI’s Model view to define how the four tables connect. The data model follows a star-schema pattern: a central fact table (Sales Transactions) surrounded by dimension tables that provide context.

From Table	To Table	Relationship Column
CityTier (City)	PinCode-Geo (City)	**City**
PinCode-Geo (PinCode)	Sales Transactions (PinCode)	**PinCode**
Product Mapping (Product_Code)	Sales Transactions (Product_Code)	**Product_Code**

This chain of relationships means that a single filter — say, selecting “Tier 1” in a slicer — propagates automatically from CityTier through PinCode-Geo into the Sales Transactions fact table. Every visual on the dashboard updates simultaneously without any additional configuration.

5. Calculated Columns & Business Metrics

5.1 Net Units: Accounting for Cancellations

Gross units sold is a misleading metric in retail — it overstates actual volume by including orders that were later cancelled. RetailMax’s finance team needs Net Units as the authoritative measure of fulfilled demand.

Calculated Column: Net_Units

Net_Units = Units - Cancelled_Units This column is added directly to the Sales Transactions table in Power Query. Every revenue and quantity calculation in the model flows from Net_Units, ensuring that cancelled orders are never double-counted in business reports.

5.2 Day-of-Week Analysis: Understanding Demand Rhythms

The operations team wants to understand whether RetailMax sees higher demand on weekdays versus weekends. This insight directly informs staffing levels, marketing campaign timing, and flash sale scheduling. A calculated column OrderDayOfWeek is added to capture the day name for every transaction.

Calculated Column: OrderDayOfWeek

OrderDayOfWeek = FORMAT(Order_Date, "dddd") This returns values like Monday, Tuesday, … Sunday for each transaction row. When plotted on a bar chart ordered by weekday, it immediately reveals peak trading days — a pattern invisible in the raw transaction data.

5.3 Week Start: Tracking Weekly Performance

In addition to daily patterns, the sales head needs to track rolling weekly performance — comparing this week vs. last week vs. the same week last year. This requires normalizing every transaction date to the start of its calendar week (Monday).

Calculated Column: OrderWeekStart

OrderWeekStart = Order_Date - WEEKDAY(Order_Date, 2) + 1 Weekday mode 2 makes Monday = 1, so subtracting (weekday − 1) always lands on the Monday of that week. Grouping by OrderWeekStart gives the analytics team clean week-over-week trend lines.

6. KPI Framework & DAX Measures

6.1 Core Business Metrics

The following KPIs are defined, agreed upon by stakeholders, and built as DAX measures in Power BI. Using measures (rather than calculated columns) ensures they respond dynamically to every slicer, filter, and drill-down in the dashboard.

KPI / Metric	Business Definition	DAX / Power Query Logic
Total Revenue	Sum of all revenue from fulfilled orders	`SUM(Sales[Revenue])`
Net Units Sold	Total fulfilled units after cancellations	`SUM(Sales[Net_Units])`
Cancellation Rate	% of gross units that were cancelled	`DIVIDE(SUM(Sales[Cancelled_Units]), SUM(Sales[Units]))`
Unique Customers	Count of distinct customer IDs in period	`DISTINCTCOUNT(Sales[Customer_ID])`

Total Transactions	Count of individual order lines	`COUNTROWS(Sales)`
Avg Revenue / Transaction	Revenue efficiency per order	`DIVIDE([Total Revenue], [Total Transactions])`
Revenue by Zone	Revenue split by geographic zone	`CALCULATE([Total Revenue], ALLEXCEPT(CityTier, CityTier[Zone]))`
WoW Revenue Growth	Week-on-week revenue change %	`DIVIDE([This Week Revenue] - [Last Week Revenue], [Last Week Revenue])`

7. Reports & Analytical Views

7.1 Report Design Philosophy

Rather than building one enormous report that tries to answer every question, the analytics team designs a suite of focused analytical views — each answering a specific business question for a specific audience. All views share the same underlying data model and are accessible from a single Power BI workspace.

Revenue Performance Report

Audience: VP Finance, CEO

- Total and Net Revenue trend lines by month, quarter, and year
- Revenue breakdown by Zone, City Tier, and individual City
- Revenue contribution by product category and sub-category
- Year-over-Year and Month-over-Month growth comparison

Cancellation & Loss Analysis Report

Audience: Operations Manager, VP Finance

- Cancellation Rate tracked over time — identifying worsening trends before they become P&L issues
- Cancellations broken down by product category to identify chronic problem SKUs
- Cancellation heatmap by city and zone to flag regional fulfilment failures
- Gross vs. Net Units comparison chart for management visibility

Customer & Demand Trends Report

Audience: Sales Head, Marketing

- Unique customer count by month — distinguishing new vs. returning customer trends
- Weekday vs. weekend demand chart (using OrderDayOfWeek) to optimize promotional timing
- Weekly trend lines (using OrderWeekStart) for operational rhythm tracking
- Customer concentration by city tier to guide acquisition strategy

Product Performance Report

Audience: Sales Head, Category Managers

- Revenue and units by product category, sub-category, and SKU
- Top 10 and Bottom 10 products by net revenue and cancellation rate
- Product performance drill-through by zone and city tier

8. Executive Management Dashboard

8.1 Dashboard Purpose

The management dashboard is the single-page, always-on view that leadership uses daily. It is not a detailed analytical report — it is a command centre designed to surface exceptions, confirm performance against targets, and trigger deeper investigation when needed.

8.2 Dashboard Layout & Key Visuals

Dashboard Zone	Visual Type	Business Question Answered
Top Row — KPI Cards	Card visuals with trend arrows	Are revenue, units, customers, and transactions on track this period?
Centre Left — Revenue Trend	Line chart (monthly)	Is revenue growing, stable, or declining?
Centre Right — Zone Map	Filled map / bar chart	Which zones are the strongest and weakest performers?
Bottom Left — Weekly Trend	Bar/line chart by OrderWeekStart	How does this week compare to prior weeks?
Bottom Centre — Day Heatmap	Column chart by OrderDayOfWeek	Which days drive peak volume?
Bottom Right — Top Products	Table or bar chart	What are the highest and lowest revenue products?
Right Panel — Slicers	Dropdown/button slicers	Allow filtering entire dashboard by Date, Zone, City Tier, Category

8.3 Interactivity & Self-Service

The dashboard is designed for self-service exploration. Every visual acts as a cross-filter: clicking on a zone in the map automatically filters the weekly trend, KPI cards, and product table to show only that zone's data. Drill-through pages allow a manager to right-click any city name and navigate to a full city-level analysis without leaving the dashboard.

9. Business Outcomes & Value Delivered

9.1 From Scattered Data to Unified Intelligence

Before this Power BI implementation, RetailMax's reporting cycle took 3–4 days each month as analysts manually reconciled data from multiple spreadsheets. Reports frequently contained errors due to unmatched city names and manually calculated cancellation figures.

After implementation:

- The entire data pipeline from raw files to refreshed dashboard takes under 5 minutes.
- All KPIs are calculated consistently from a single, governed data model.

- Leadership can answer ad-hoc strategic questions in real time using slicers, without requesting new reports from the analytics team.
- The cancellation rate metric — previously invisible — is now tracked weekly, enabling the operations team to intervene before fulfilment failures escalate.
- Weekday/weekend demand analysis has allowed the marketing team to shift promotional spend to Thursday–Friday, resulting in a measurable uplift in weekend conversion.

> **Key Takeaway**
> This case study demonstrates that Power BI's true value lies not in the visuals themselves, but in the disciplined data foundation beneath them. Clean data + a well-designed model + thoughtfully defined KPIs = reports that leadership actually trusts and acts on.

10. Power BI Skills Demonstrated

Skill Area	Demonstrated Through
Power Query — Data Integration	Loading multiple files, promoting headers, ensuring correct data types across all four source tables
Power Query — Data Cleansing	Filtering null Zone/CityTier records, text transformation to strip country suffix from City names
Power Query — Column Creation	Creating Net_Units as a calculated column for reliable downstream aggregations
Data Modeling	Building a star-schema model with three dimension tables linked to the fact table via City, PinCode, and Product_Code
DAX — Calculated Columns	OrderDayOfWeek (FORMAT function) and OrderWeekStart (date arithmetic with WEEKDAY) for time intelligence
DAX — Measures	Revenue, Net Units, Cancellation Rate, Unique Customers, WoW Growth using SUM, DIVIDE, DISTINCTCOUNT, COUNTROWS, and CALCULATE
Report Design	Four focused analytical reports each targeting a specific stakeholder question
Dashboard Design	Single-page executive view combining KPI cards, trend lines, geographic visuals, and cross-filtering slicers

End of Case Study